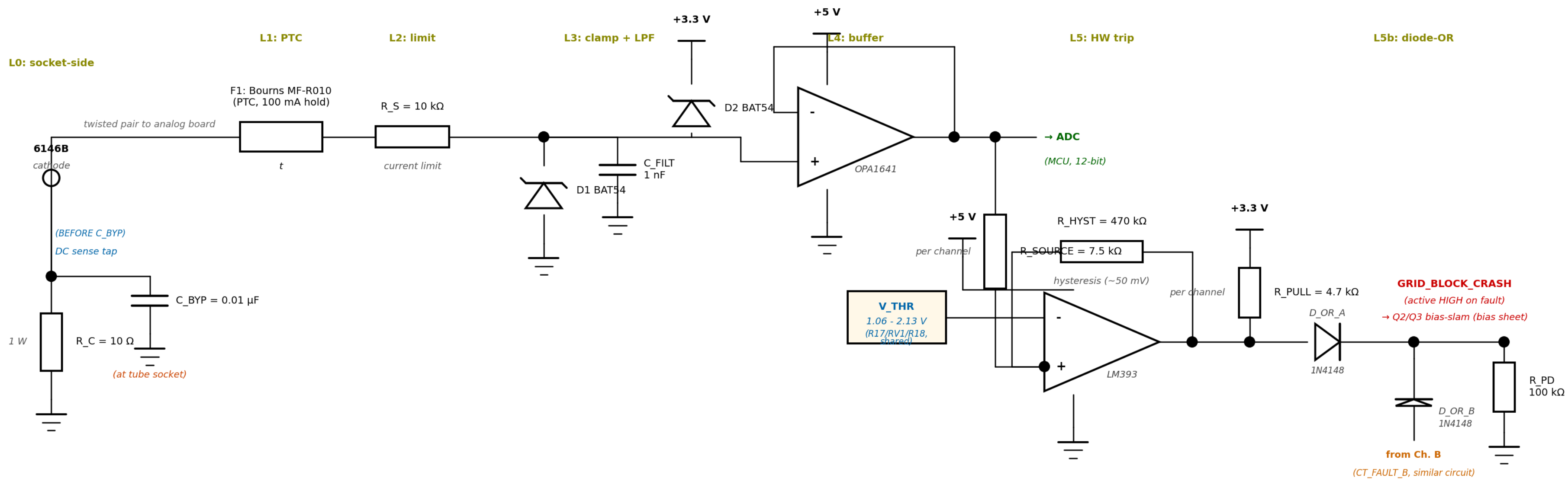
Per-Tube Cathode Current Monitor + Failsafe Chain

Seven defensive layers from cathode (+1 V nominal, may see 600 V on fault) to MCU ADC. One channel shown; ×2 for the push-pair, combined via the diode-OR on the right.



Operating point (per tube at full drive):	*I_cathode ≈ 100 mA → V_sense ≈ 1.0 V at ADC input → ~30% ADC range*
Trip threshold + hysteresis:	*V_THR (RV1 wiper) sits at ~1.5 V nominal (~150 mA cathode); R_HYST=470 kΩ feeding back through R_SOURCE=7.5 kΩ gives ~50 mV hysteresis. LM393 (+) > (-) → OC releases → fault HIGH.*
Total fault response (cathode overcurrent → tubes cut off):	*< 100 µs total -- buffer + comparator + hysteresis settling (~10 µs) + diode-OR + OPA454 slew from operating bias to -85 V rail at 13 V/µs (~2 µs); no SR latch needed.*
Diode-OR vs. wired-tie (why two channels are NOT tied to one pull-up):	*LM393 OC tied directly with single R_PULL gives wired-AND of fault (HIGH only when BOTH channels fault) -- wrong polarity. Diode-OR with per-channel R_PULL gives true OR.*
Critical layout rule:	*DC sense tap leaves R_C top BEFORE C_BYP. Three grounds (GND_RF, GND_SENSE, GND_DIG) meet at one star point on the analog board.*

Figure 1. Per-tube cathode current monitor + failsafe chain. Levels 0–5 + the L5b diode-OR combiner shown; Levels 6–7 (firmware) are described on the next page.

Defensive Layers and Bill of Materials

Level	Component(s)	Response time	Failure mode caught
L0	R_C (10 Ω, 1 W), C_BYP (0.01 µF NP0)	—	Normal sense + RF rejection at tube socket
L1	F1 = Bourns MF-R010 PTC (100 mA hold)	~ 1 s	Sustained fault thermal cooking of downstream parts
L2	R_S = 10 kΩ 1/4 W series limiter	instant	Fault current limited to 60 mA at 600 V cathode short
L3	D1, D2 = 1N5817 Schottky clamps; C_FILT = 1 nF	instant	Voltage spikes beyond ±supply rails; RF anti-alias
L4	U13, U14 = OPA1641 unity-gain buffer (+5 V / GND)	< 1 µs	ADC physically isolated; OPA1641 has ±20 V input protection
L5	U12 = LM393, V_THR = 1.06–2.13 V (R17/RV1/R18 trim), R_SOURCE = 7.5 kΩ, R_HYST = 470 kΩ (per channel)	< 10 µs	Hardware overcurrent trip → fault HIGH at LM393 OC
L5b	Diode-OR: D_OR_A, D_OR_B (1N4148) + R_PD = 100 kΩ pulldown	< 1 µs	Combine both channels → GRID_BLOCK_CRASH (active HIGH) → Q2/Q3 bias-slam on bias sheet → grid bias to –85 V → tubes cut off
L6	Firmware ADC sampling at 1 kHz on ESP32-S3 ADC1, threshold checks	~ 5 ms	Soft trip + warnings + UI alerts
L7	NVS fault log (timestamp, tube_id, type, current, duration)	persistent	Post-mortem analysis across power cycles

Bill of Materials (per tube unless noted; ×2 for the push-pair)

Ref	Part	Qty	Notes
R_C	10 Ω 1 % 1 W metal film	1/tube	Mount AT the tube socket. Vishay PR01000101000JR500 or similar.
C_BYP	0.01 µF NP0 ceramic, 100 V	1/tube	Low-ESL package, ≤5 mm leads, at socket pin
F1	Bourns MF-R010 PTC	1/tube	100 mA hold, 200 mA trip, V_max 60 V
R_S	10 kΩ 1 % 1/4 W metal film	1/tube	Fault current limit
D1, D2	1N5817 Schottky	2/tube	Clamps to GND and +3.3 V
C_FILT	1 nF X7R 50 V	1/tube	Anti-alias + RF reject
U13, U14	OPA1641 single op-amp	1/tube	Unity-gain buffer. Built-in ±20 V differential input protection.
U12	LM393 dual comparator	1	One IC handles both tubes
U11	LM4040DIZ-5.0	0	Already in grid_bias BOM. Same +5 V_REF rail powers the threshold divider.
R17 / RV1 / R18	27 kΩ / 10 kΩ 25-turn cermet / 10 kΩ	1 each	Shared threshold divider on +5 V_REF. Wiper range 1.06–2.13 V; nominal set point ~1.5 V (≈150 mA cathode trip).
R_SOURCE	7.5 kΩ 1 %	1/tube	R20 (ch.B) and R22 (ch.A). OPA1641 OUT → LM393 (+) input. Sets hysteresis with R_HYST.
R_HYST	470 kΩ 1 %	1/tube	R21 (ch.B) and R23 (ch.A). LM393 OUT → (+) input. ~50 mV hysteresis.
R_PULL	4.7 kΩ 1 %	1/tube	Pull-up on each LM393 OC output to +3.3 V (NOT shared — see diode-OR below).

D_OR_A, D_OR_B	1N4148 (or 1N5817 from stock)	1/tube	Diode-OR combiner. Anode at LM393 OC, cathodes tied at GRID_BLOCK_CRASH node.
R_PD	100 kΩ 1 %	1	Pulldown at the diode-OR common node — defines LOW when no fault.
C_DEC	100 nF X7R + 10 µF bulk	several	Decoupling at OPA1641, LM393, LM4040 supply pins.

Operating Points and Trip Levels

Condition	I_cathode	V_sense	ADC count (12-bit)	Action
Tubes cut off (key-up)	0 mA	0 V	0	normal idle
Nominal operating	100 mA	1.0 V	~1240	normal log
Soft warning threshold	120 mA	1.2 V	~1490	Firmware: warn UI, reduce envelope by 10 %
Hard fault (trip)	~150 mA	~1.5 V	~1860	LM393 fault HIGH → diode-OR → GRID_BLOCK_CRASH → Q2/Q3 bias-slam → grid at –85 V
Sustained overcurrent	200 mA+	2.0 V+	~2480+	PTC F1 also opens within ~1 s

Design Rationale and Procedures

Why grid-bias slam (not screen, plate, or filament)?

Plate: 600 V / 250 mA. Hard to switch reliably; relays slow, MOSFETs expensive at that V_DS. Capacitor discharge dumps significant energy.

Screen: workable (low current, fast off) but adds a dedicated switching MOSFET and a separate failure surface. Not re-used by any other subsystem.

Filament: cathode thermal mass means seconds to cool. No protection against an in-progress fault.

Grid bias (chosen): the OPA454 grid-bias generators on the bias sheet are already there for normal operation. A single 2N7000 per tube (Q2, Q3) sits across each OPA454 summing junction. When GRID_BLOCK_CRASH goes HIGH, Q2/Q3 short the summing junctions to GND; the OPA454 slams its output to the –90 V rail at 13 V/µs. Both tubes go to deep cutoff (~–85 V grid) in under 100 µs. No screen-voltage switch, no extra high-voltage MOSFETs, no firmware in the fast path.

Calibration Procedure

1. Offset zero: Tubes in deep cutoff (key-up). Read each tube's ADC count; store as cathode_offset[tube_id].

2. Gain calibration: Insert a known shunt resistor in series with the cathode return, with a DMM measuring I_cathode. Drive the PA to multiple current levels (50, 75, 100, 125 mA). Record ADC count at each. Fit a line: I_cathode_mA = (ADC_count – offset) × gain. Store gain[tube_id].

3. Comparator threshold trim: Drive the tube to I_cathode = 150 mA. Adjust RV1 (10 kΩ 25-turn cermet between R17 and R18) until LM393 just trips. Verify it un-trips at 100 mA (hysteresis check). Store the trim date in NVS as trip_calibration_date.

4. Verification: Trip-test with the bias-slam path live. Verify the grid bias swings from its operating point (~–60 V) to the –85 V rail within 100 µs of the LM393 trip event. Use a scope on a grid pin (with HV probe) to time-resolve the response.

Re-run calibration every 100 operating hours or after any servicing.

Layout and Grounding Rules

• **R_C and C_BYP mount AT the tube socket cathode pin**, with leads ≤ 5 mm. C_BYP's ground end goes to local chassis (GND_RF).

• **DC sense tap leaves R_C top BEFORE C_BYP.** If reversed, 14 MHz RF currents flow through the sense path instead of through C_BYP — sense becomes useless.

• **Sense wire is twisted pair**: hot signal + analog GND return. Common-mode pickup ≈ 0. Optionally shielded with shield grounded at the analog-board end only.

• **Three separate "grounds":** GND_RF (tube socket area, bypass caps); GND_SENSE (analog board signal); GND_DIG (MCU board). All three converge at exactly one star point on the analog board. No daisy-chain.

• **Op-amp and comparator on analog board**, NOT at the tube socket. Keeps them within their thermal envelope and away from the RF area.

• **+3.3 V rail to which D2 clamps**: needs ≥ 10 µF bulk cap near the clamp node to absorb fault energy without sagging.

Open Items

ADC choice — RESOLVED 2026-06-23: built-in ESP32-S3 ADC1, 12-bit. Two channels (I_CATHODE_A, I_CATHODE_B) tap the OPA1641 outputs before R_SOURCE and route to ADC1-capable Metro pins.

SR latch — RESOLVED 2026-06-23: no latch chip. The diode-OR + bias-slam path is unlatched; the slam holds as long as cathode current is above threshold (with ~50 mV hysteresis). Firmware logs the event from the ADC path. Cleanup of one part if needed: a firmware-asserted CLEAR GPIO that pulls the diode-OR common node LOW through a small resistor.

1N5817 current margin: 60 mA F1-limited fault is well within the 1 A continuous / 25 A surge rating. No upsizing needed.

Remaining bias-sheet work: add the diode-OR diodes (D_OR_A, D_OR_B), second R_PULL, and R_PD = 100 kΩ pulldown on bias.kicad_sch; add I_CATHODE_A / I_CATHODE_B hierarchical labels routing to the Metro ADC1 pins on the arduino sheet.

Per-pair sum monitor: optional third comparator with threshold at 280 mA total cathode current (sum of both tubes), to catch the case where both tubes drift hot together without individual tube exceeding its threshold.